

Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibility	Email				
Eloi Iradukunda	PCA implementation (covariance, eigenvalues/eigenvectors), data preprocessing, report writing	e.iradukund2@alustudent.com				
Innocent Tito Muvunyi	Data collection (African dataset), handling missing values & non-numeric encoding, visualization support	i.muvunyi@alustudent.com				
[Name]						
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Task Allocation and Tracker						
	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Dataset selection (African dataset with NaNs + non-numeric column)	Innocent Tito Muvunyi	02-06-2026	Completed	Yes	Chosen dataset with missing values and categorical field
	Data cleaning (imputation + encoding non-numeric data)	Innocent Tito Muvunyi	03/06/2026	Completed	Yes	Handled missing values and categorical encoding
	Covariance matrix computation	Eloi Iradukunda	04/06/2026	Completed	Yes	Implemented using NumPy from scratch
	Eigenvalues & eigenvectors computation	Eloi Iradukunda	05/06/2026	Completed	Yes	Manual decomposition without sklearn
	Sorting eigenvalues & selecting principal components	Eloi Iradukunda	07/06/2026	Completed	Yes	Explained variance used for selection
	PCA projection implementation and Explained variance analysis & justification	Eloi Iradukunda	08/06/2026	Completed	Yes	Data projected onto principal components and tradeoff between dimensionality and information loss explained
	Visualization (before & after PCA plots)	Innocent Tito Muvunyi	09/06/2026	Completed	Yes	Matplotlib used for clear comparison plots
	Performance optimization for large dataset	Innocent Tito Muvunyi	09/06/2026	Completed	Yes	Vectorized NumPy operations applied
	Final notebook review & GitHub upload	Both	10/06/2026	Completed	Yes	Final cleanup and submission preparation
Meeting Attendance Log						
Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
02/06/2026	Dataset selection & project setup	[Name]	Eloi and Innocent	Chose African dataset with missing values and categorical columns, agreed on PCA approach without sklearn	Download dataset and prepare initial cleaning plan	
03/06/2026	Data preprocessing strategy	[Name]	Eloi and Innocent	Discussed handling NaNs, encoding categorical variables, and normalization approach	Clean dataset and prepare covariance structure	
05/06/2026	PCA implementation planning	[Name]	Eloi and Innocent	Defined steps: covariance matrix - eigen decomposition - sorting eigenvalues - projection	Start coding PCA from scratch in NumPy	
07/06/2026	results review & visualization	[Name]	Eloi and Innocent	Reviewed PCA outputs, explained variance results, and visualization structure	Finalize plots (before vs after PCA)	
09/06/2026	Final integration & submission prep	[Name]	Eloi and Innocent	Checked notebook completeness, GitHub structure, and report formatting	Upload final notebook + dataset + PDF task sheet	